Mediator

Chat room without user-to-user coupling

Season 3 · GoF · Beginner

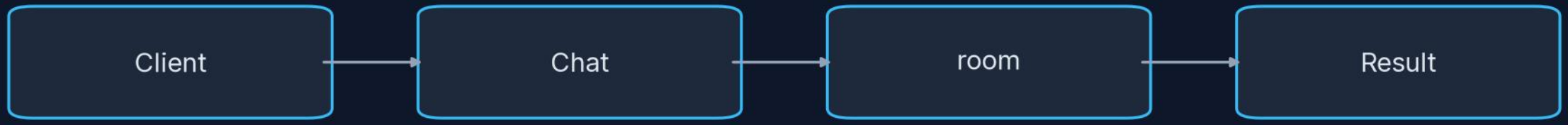
Today's Hook

- Problem: Chat room without user-to-user coupling
- Naive pain cue: caught=missing:Cara
- We will refactor to Mediator with a runnable Demo.
- Failure path is locked by ProblemTest.

The Problem

- Scenario: Chat room without user-to-user coupling
- Constraints: solve «Chat room without user-to-user coupling» with Mediator — keep the demo runnable and the failure path observable.
- Naive pain: Naive version fails to cleanly solve: Chat room without user-to-user coupling
- Success looks like: Ada→Bob:hi

Structure



Mediator: Chat room without user-to-user coupling

Real-World Scenario

- Running example: Chat room without user-to-user coupling
- Key types: ChatRoom
- Happy-path stdout cue: Ada→Bob:hi
- Failure-path stdout cue: caught=missing:Cara

Naive Code

```
package com.buildwithwaqar.patterns.gof.behavioral.mediator.problem;

import java.util.*;

public final class NaiveChat {
  public static void dm(String from, String to, String msg){ System.out.println(from+"->"+to+":"+msg); }
}
```

Why Naive Hurts

- Naive version fails to cleanly solve: Chat room without user-to-user coupling
- Without Mediator, call sites absorb the wrong details.
- Failure mode to show on camera: caught=missing:Cara
- New requirements force shotgun edits across the naive code.

Solution

```
package com.buildwithwaqar.patterns.gof.behavioral.mediator.problem;

import java.util.*;
import java.util.concurrent.*;
import java.util.concurrent.atomic.*;
import java.util.function.*;

final class ChatRoom {
  private final Map<String, Consumer<String>> users = new HashMap<>();
  private final List<String> log = new ArrayList<>();
  void join(String name, Consumer<String> inbox){ users.put(name, inbox); }
  void send(String from, String to, String msg){
    var inbox = users.get(to);
    if (inbox==null) throw new IllegalArgumentException("missing:"+to);
    String line = from+"->"+to+":"+msg; log.add(line); inbox.accept(line);
  }
  List<String> log(){ return List.copyOf(log); }
}
```

Runnable main

```
package com.buildwithwaqar.patterns.gof.behavioral.mediator.problem;
import java.util.*; import java.util.concurrent.*;
public final class ProblemDemo {
  public static void main(String[] args) throws Exception {
    System.out.println("=== Mediator: chat room ===");
    System.out.println("--- naive ---"); NaiveChat.dm("Ada","Bob","hi");
System.out.println("--- mediator ---");
var room=new ChatRoom(); List<String> bob=new ArrayList<>();
room.join("Ada", m->{}); room.join("Bob", bob::add);
room.send("Ada","Bob","hi");
System.out.println("bobInbox="+bob); System.out.println("log="+room.log());
try { room.send("Ada","Cara","x"); } catch (IllegalArgumentException e){ System.out.println("caught="+e.getMessage()); }
  }
}
```

Walkthrough Output

```
=== Mediator: chat room ===
--- naive ---
Ada->Bob:hi
--- mediator ---
bobInbox=[Ada->Bob:hi]
log=[Ada->Bob:hi]
caught=missing:Cara
```

Step-by-Step Trace

- Match stdout to stages — look for: Ada→Bob:hi
- Then the injected failure: caught=missing:Cara
- Re-run: java -cp ... 19-mediator ProblemDemo (see episode folder).
- Pause the video and predict the next printed line.

Modern Java 25 / Zazr

```
// Java 25 direction for Mediator
// Prefer records + sealed where they shrink the demo.
// Problem: Chat room without user-to-user coupling
```

Trade-offs

- Use Mediator when «Chat room without user-to-user coupling» repeats and a simpler feature does not erase it.
- Refuse extra types when a sealed switch or a function value already suffices.
- Teaching demos omit production metrics, retries, and multi-region clocks.
- Interview soundbite: state the force behind Mediator in product language.

Related Patterns

- Name the pattern usually confused with Mediator and separate the forces.
- Sketch the roles for «Chat room without user-to-user coupling» from memory in sixty seconds.
- Point to the next episode in this season on the playlist map.
- Link this episode's series/ folder in the YouTube description.

Recap

- Problem: Chat room without user-to-user coupling
- Naive pain → Mediator boundary → Demo stdout.
- Remember the failure cue: caught=missing:Cara
- Break a ProblemTest assertion, watch it fail, restore it.